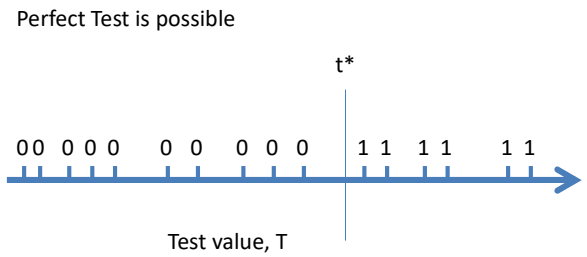
ROC CURVES: AN ALTERNATIVE ASSESSMENT OF A MODEL OR DIAGNOSTIC

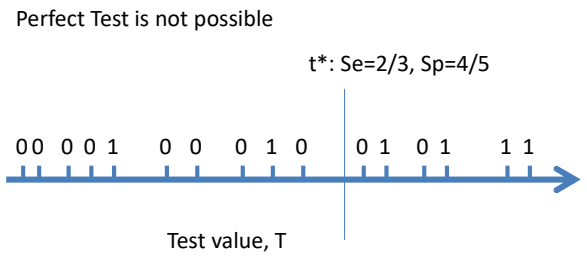
In disease data, assessing predictive quality using ROC curves (Receiver Operating Characteristic curves) is an alternative to the use of accuracy and bias. Suppose you have a disease where individuals are either diseased (denoted 1) or healthy (denoted 0), and that there is a diagnostic test for this disease which gives a quantitative value, T. Then if my test is perfect there will be a value t* such that if the individual is diseased then T>t*, and if it is healthy then T≤ t*. However diagnostic tests are rarely perfect and imperfections are modelled by epidemiologists in terms of two variables Sensitivity (Se) and Specificity (Sp). The Se is the probability that T> t* given that the individual is diseased i.e. what fraction of the truly diseased individuals are identified as being diseased by the test. The Sp is the probability that T≤ t* given that the individual is healthy i.e. what fraction of the truly healthy individuals are identified as being healthy by the test.

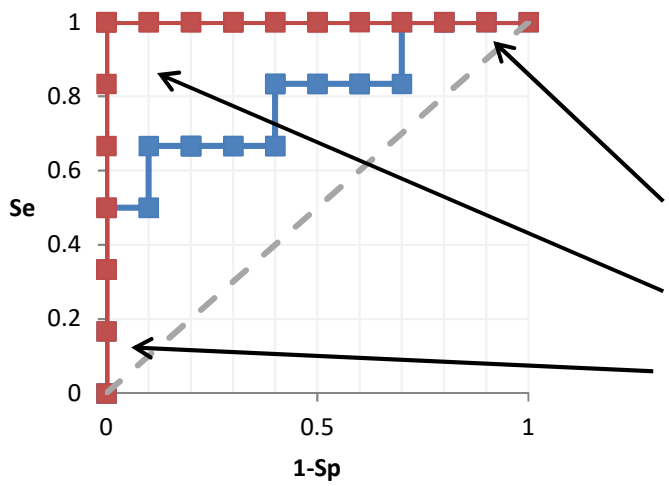
We can look at the test performance as a function of the threshold value of t that we choose to define diseased and healthy. Suppose that the test has the potential to be perfect, this is shown in the diagram below.



Now calculate sensitivity and specificity as functions of t, Se(t) and Sp(t). If t is very large, then all values of T for both healthy and diseased will be < t and all animals will be declared healthy. Then Sp will be 1 but Se will be 0. Imagine t decreasing towards t* then the test will always declare the healthy animals as healthy, but as t approaches t* more and more of the diseased animals will be declared as being diseased so the Se will increase until t is less than the lowest possible value of the diseased individuals then Se will be 1. So at t = t*, both Se=1 and Sp=1, and the test is perfect. If t is then deceased below t*, Se remains 1 as all diseased animals are classified as diseased, but now more and more healthy animals are being classified as diseased so Sp decreases below 1. When t is very low, then all animals will be classified as diseased and Sp=0 even though Se=1. If Se is plotted against 1-Sp for different values of t we obtain a step function with 1 step, which has an area underneath it of 1. The actual test adopted will have any value t* where Se = Sp = 1.

Suppose the test value had no potential to be perfect – there is overlap in the distribution of T values for the healthy and diseased. The quality of the test will depend on these distributions around the region of the overlap. Look at the figure below. Now when t is large, Se = 0 and Sp =1, as before, but as t decreases Se and Sp have values where both are less than 1. At the old t* value neither Se nor Sp are 1, and (here) Se= 2/3 and Sp =4/5. Now the plot of Se against 1-Sp is a step function with several steps, and the area under the curve (typically denoted AUC) is 0.8 < 1. Clearly the closer to the perfect diagnostic and AUC=1 the better.





ROC curves for the perfect test (Red) and the imperfect test (Blue). The dashed grey line is the line for a random test. (Red and Blue lines coincide for extreme values of the threshold.)

Very low threshold values of t

Intermediate threshold values of t

Very high threshold values of t

If the data were much larger and the test T is continuous then the step function for an imperfect test will become smoother, as there are many more steps. If the test were completely random and had no use as a diagnostic test, the distributions of T for healthy and diseased individuals would be identical and in this case (assuming the data is large) Se = 1- Sp for all values of t; i.e. as Se = Prob(T>t | Diseased) = Prob(T>t | Healthy) = 1 – Prob(T≤t | Healthy) = 1- Sp as the distributions for healthy and diseased are the same. Then the area under the curve is 0.5.

In practice the threshold value is determined by the consequent actions, but the ROC curve summarises the possibilities. If it is (say) a medical screening test where the action associated with the test being 'Diseased' is simply further more accurate tests then it is appropriate for a very high Se and a lower Sp. In livestock disease, the consequence may be to cull an animal and pay compensation (as in bovine TB in UK), then the decision may be to lower Se and ensure a high Sp.

The relevance to linear genetic models is straightforward. From our analyses we will typically produce a genetic predictor T_G as a measure of genetic risk of developing the disease. So we can construct a ROC curve by considering what happens as we change the threshold. The best model would give us the best AUC. Since we are only considering the genetic part of the risk then phenotypic disease data will have an additional environmental component that influences the development of disease and the observed phenotypes. Therefore unless the genetic component completely determines the development of disease the AUC will never be 1, and as the genetic component becomes smaller (i.e. heritability decreases) so maximum possible AUC from T_G, i.e. assuming we are perfectly predicting the genetic component, will shrink towards 0.5. For more information see http://gump.qimr.edu.au/genroc/.

Therefore to test a genetic model for the disease we may construct a predictor from a training set, and then validate it using the AUC of the ROC curve in a completely independent sample, the validation set. This is an alternative to correlating the disease phenotypes (0's and 1's) in the validation set with their predictors. Looking at the figure of the perfect test it is clear that the correlation of the 0/1 values with T is not 1; as variation in T within the 1's and variation in T within the 0 do not affect the 0/1 scores – although it may be reflecting a true but unmeasured continuous risk function.

One of the benefits of the ROC approach is that Se and Sp are independent of how the data was gathered, providing the 'Diseased' and 'Healthy' sets are random samples of their respective populations. Se and Sp are calculated within their respective groups for any threshold. This is in contrast to the correlation between the prediction and disease status, as this will depend on the selection policy: is this a random sample of the population; has it been purposely collected as a case-control; if case-control what was the split between the two groups; what is the prevalence of the disease in the population.

BEWARE!

I have described ROC curves using Se and Sp, which are well-defined, and without using phrases such as e.g. 'true positive *rate*'. Unless used with care such 'rate' phrases are ambiguous!

Construct the 2x2 table. The labels TP, FN, FP, TN are (I think) accepted usage.

	Test 'Disease'	Test 'Healthy'
Disease	True Positive (TP)	False Negative (FN)
Healthy	False Positive (FP)	True Negative (TN)

The word 'rate' is ambiguous: e.g. is 'true positive rate' TP/(TP+FP), where the rate is per positive test, or is it TP/(TP+FN), where the rate is per case? Both rates make sense when used in their appropriate context! For example, in giving guidance to those with a positive screening test, it may be important to tell the patient that the particular test used has a high 'false positive rate' defined by FP/(TP+FP); i.e. the probability that there is no disease is high despite the test result. This 'false positive rate' cannot be determined from Se and Sp without knowing the prevalence.

The standard 'rate' parameters used in epidemiology are Se and Sp, and are defined by the true status of an individual, **not** the test outcome: i.e. Se = TP/(TP+FN) and Sp = TN/(FP+TN).

If the word 'rate' is used, make sure it is defined!